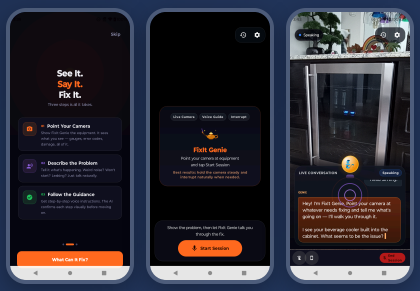
Gemini Live Agent Challenge · The Live Agent Category

FixIt Genie

See. Hear. Fix. — Your AI repair genie that sees through your camera and talks you through the fix.

Google ADK | Gemini 2.5 Flash Native Audio | Google Cloud Run | Cloud Firestore Vector Search | Ray-Ban Meta Glasses | Native Android



THE PROBLEM

Equipment fails. Guidance is blind.

When something breaks, people reach for a PDF manual, a search engine, or a YouTube video that never quite matches the exact model in front of them. None of these tools can *see what you're looking at*. A maintenance tech inside an electrical panel can't hold a phone. A mechanic with greasy hands can't type a query. Traditional tools fail exactly when the stakes are highest.

THE SOLUTION

An AI that sees, hears, and guides — in real time.

FixIt Genie streams live camera frames and microphone audio to a Gemini 2.5 Flash Native Audio agent via Google ADK. The agent identifies equipment visually, understands natural speech, retrieves domain knowledge, and talks the user through the fix — step by step, hands-free, interruptible at any moment. Built on Google Cloud Run with Firestore vector search and four function tools.

1 FPS	16 kHz	4	302
Live camera stream to agent	Bidirectional PCM audio	Function tools registered with ADK agent	Automated tests (129 backend + 173 Android)

Live Agent Capabilities

Interruptible
Say "wait" or "stop" mid-response. Gemini's native VAD detects speech, sends `interrupted:true`, agent stops instantly.

Bidi Streaming
OkHttp WebSocket + ADK LiveRequest/LiveEvent. Audio and video stream up; audio + transcripts stream back — simultaneously.

Visual Awareness
Agent identifies equipment, reads error codes and gauges, and calls out what it notices — without you naming anything first.

Hands-Free
Ray-Ban Meta glasses as a live camera source. One tap switches to glasses — same agent, same pipeline, no session restart.

Safety-First
`get_safety_warnings()` is called before any physical action. Enforced in system prompt and verified by unit tests.

Domain Coverage

AUTOMOTIVE
Engine oil (P0520–P0524), battery/electrical (P0562–P0621), cooling system (P0115–P0128). Reads oil level gauges visually.

ELECTRICAL
Breaker panel reset, GFCI outlet troubleshooting. Safety warnings always issued before any panel interaction.

APPLIANCES
Washing machine (E1–E4, UE, OE), dishwasher (E15, E24), LG refrigerator (Er IF, Er FF, Er CF + 33 more codes).

System Architecture

Core Data Flow

1. **Capture** — CameraX 1 FPS JPEG + AudioRecord 16kHz PCM captured on device
2. **Stream** — OkHttp WebSocket sends ADK LiveRequest — video as JSON, audio as binary blobs
3. **Understand** — Gemini 2.5 Flash Native Audio processes multimodal stream via bidiGenerateContent
4. **Ground** — ADK tools: Firestore vector search (`find_nearest()`) + keyword fallback for equipment knowledge
5. **Respond** — 24kHz PCM audio + transcript events streamed back to app in real time

Knowledge & Grounding Stack

Firestore Vector Search — `lookup_equipment_knowledge`

Equipment documents embedded with `gemini-embedding-001` (1536-dim COSINE index) via `seed_knowledge.py` . Semantic retrieval via `find_nearest()` — "engine oil pressure alarm" matches oil system doc with zero keyword overlap.

Semantic Retrieval

Domain Skill Packages — `skills/`

3 repair domains structured as skill packages: `automotive/` , `electrical/` , `appliances/` . Each contains a `SKILL.md` with behavioral guidance and `references/` markdown docs. Content seeded into Firestore KB as the source of truth.

Domain Knowledge

Embedded Python Dict Fallback — `_KNOWLEDGE_BASE`

Fallback

In-memory keyword fallback in `tools.py` — `lookup_equipment_knowledge` falls back here if Firestore is unavailable or returns no match. Also enables unit tests with zero cloud dependencies.

Key Components

ANDROID CLIENT (KOTLIN + JETPACK COMPOSE)

SessionViewModel	Orchestrates camera, audio, WebSocket, transcript via `StateFlow<SessionUiState>`
AgentWebSocket	OkHttp WebSocket — ADK LiveRequest/LiveEvent protocol. Handles reconnect + session lifecycle
AudioStreamManager	AudioRecord 16kHz in, AudioTrack 24kHz out with hardware echo cancellation (`USAGE_VOICE_COMMUNICATION`)
GlassesCameraManager	Meta DAT SDK v0.4.0 — I420→JPEG, live toggle via `CameraSource` enum

BACKEND AGENT (GOOGLE ADK ON CLOUD RUN)

agent.py	ADK Agent with `gemini-2.5-flash-native-audio-latest` , 4 function tools, 8 telemetry callbacks
tools.py	All function tools + Firestore vector client + embedded KB fallback
deploy.sh	IaC: enables APIs, creates Artifact Registry, builds via Cloud Build, deploys with session affinity
Dockerfile	`python:3.12-slim` → `adk web --port 8080` — exposes `/run_live` WebSocket endpoint

Judging Criteria Alignment

Innovation & Multimodal UX — 40%

- ✓ Breaks the text-box paradigm — simultaneous live camera + voice + real-time spoken response
- ✓ Native interruptibility: say "wait" mid-response, agent stops immediately via native VAD
- ✓ Agent sees equipment and reads error codes without being told — trust built before guidance
- ✓ Ray-Ban Meta glasses for hands-free — changes the form factor entirely for professional use
- ✓ Animated genie avatar with audio-level ripple rings; chat transcript layout; onboarding "See It. Say It. Fix It."

Technical Implementation — 30%

- ✓ Google ADK with 4 function tools + 8 telemetry callbacks; full bidi-streaming via `adk web /run_live`
- ✓ Cloud Run with session affinity for persistent WebSocket; IaC via `deploy.sh`
- ✓ Firestore vector search (1536-dim COSINE) + embedded KB fallback — grounded, not hallucinated
- ✓ 302 automated tests (129 backend + 173 Android); safety warnings enforced and unit-tested
- ✓ Graceful error handling: reconnect on drop, embedded KB fallback, fallback tool chain

Demo & Presentation — 30%

- ✓ <4-min demo video: real-time multimodal session, interruption, tool calls in flight
- ✓ Architecture diagram with Cloud Run deployment proof
- ✓ Live backend: fixitbuddy-agent-hybxqwgczq-uc.a.run.app
- ✓ Public GitHub repo with spin-up instructions and `dev.sh` one-command local run
- ✓ Technical blog post at fixit-genie.web.app/blog.html

Tech Stack

Layer	Technology	Role
AI / Model	`gemini-2.5-flash-native-audio-latest` — bidi-streaming with custom function tools; `gemini-2.5-flash` for tool summarization; `gemini-embedding-001` 1536-dim for vector search	Core
Agent Runtime	Google ADK (`adk web`) — 4 function tools, 8 telemetry callbacks, LiveRequest/LiveEvent bidi-streaming protocol, session affinity on Cloud Run	Core
Android App	Kotlin 2.3 · Jetpack Compose + Material 3 · CameraX 1.4.1 · AudioRecord/AudioTrack · OkHttp WebSocket · Hilt 2.59.2 (KSP) · AGP 9.0.1	Client
Glasses	Meta DAT SDK v0.4.0 (`mwdat-core` , `mwdat-camera`) — I420→JPEG frames, same WebSocket pipeline as phone camera, minSdk 31	Input
Infrastructure	Google Cloud Run (2 vCPU, 2 GiB, session affinity) · Cloud Build · Artifact Registry · Cloud Firestore · Firebase Hosting (demo site)	Cloud
Long-Tail Tools	`youtube-transcript-api` (deterministic, works for any video) · `pypdf` for manufacturer PDF extraction · `requests` for manual retrieval	Tools

Google Cloud Services

☁️ **Cloud Run**
ADK agent host with session affinity for persistent WebSocket

🔥 **Cloud Firestore**
Vector search knowledge base (gemini-embedding-001, COSINE index)

🏗️ **Cloud Build**
Container builds via deploy.sh IaC — fully automated

📦 **Artifact Registry**
Docker image storage for Cloud Run deployments

✨ **Gemini API**
Native audio bidi-stream, embeddings, supporting tool calls

Bonus Points & Links

✓ **Blog Post (+0.6)**
fixit-genie.web.app/blog.html

✓ **IaC Deployment (+0.2)**
backend/deploy.sh — full GCP setup in one command

Resource	URL
GitHub	github.com/dabra-labs/fixIt-genie
Live Demo Site	fixit-genie.web.app
Technical Blog	fixit-genie.web.app/blog.html
Live Backend	fixitbuddy-agent-hybxqwgczq-uc.a.run.app
Deployment Proof	github.com/dabra-labs/fixIt-genie/blob/main/DEPLOYMENT.md

THE PATTERN GENERALIZES

New vertical = new tools + new system prompt. The camera/audio pipeline, WebSocket protocol, ADK skills structure, safety-first design, and glasses support are reusable infrastructure. Swap the knowledge tools — the streaming core delivers the same conversational, visually-grounded experience.

VERTICAL OPPORTUNITIES

Industrial maintenance — factory floor, Ray-Ban glasses, OEM manuals, hands-free
Fleet service — truck VIN, active TSBs, roadside repair escalation
Healthcare equipment — biomedical tech, FDA databases, radiation safety warnings
Insurance claims — damage assessment, repair cost grounding, structured reports